Title Slide

Ensemble Models vs Single Models

- Fair comparisons across many datasets
- Focus on classification and regression
- Goal: evidence, not assumption

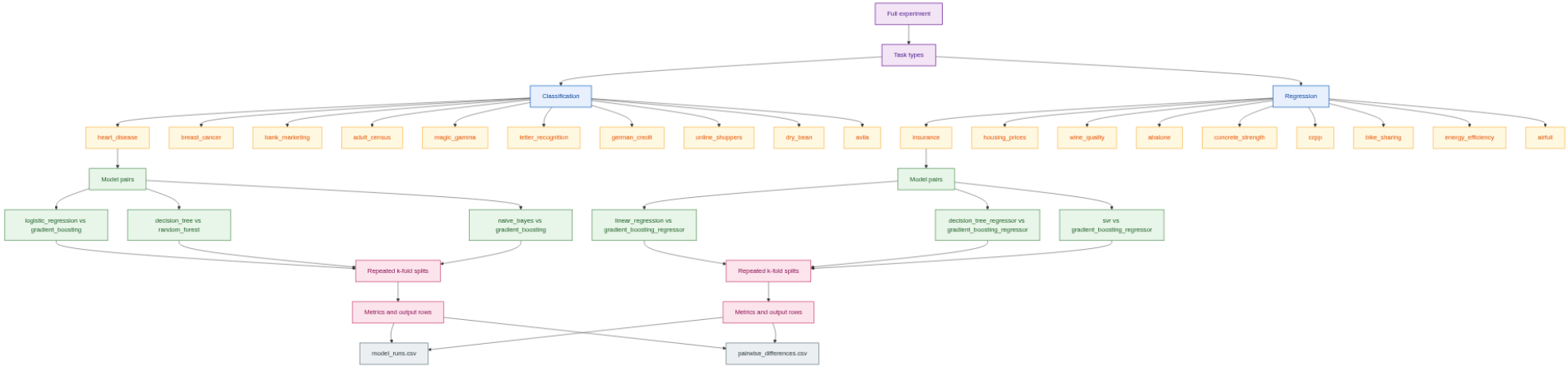
Problem Statement

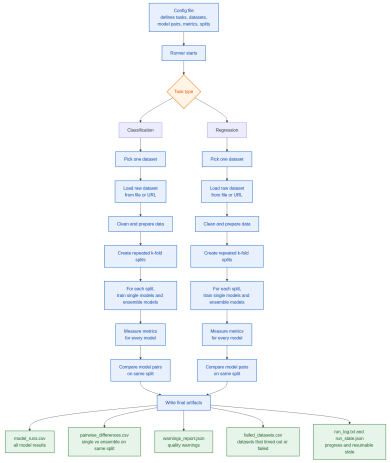
- Do ensemble models consistently beat single models?
- Does result change by dataset?
- Does result change by metric?
- Can we defend answer with paired evidence?

Objectives

- Build fair, matched comparisons
- Store raw results and paired differences
- Summarize patterns from several angles
- Reach defensible conclusion without overclaiming

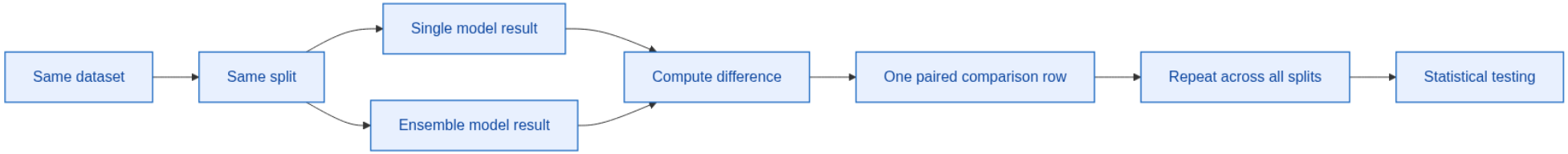
Data Collection





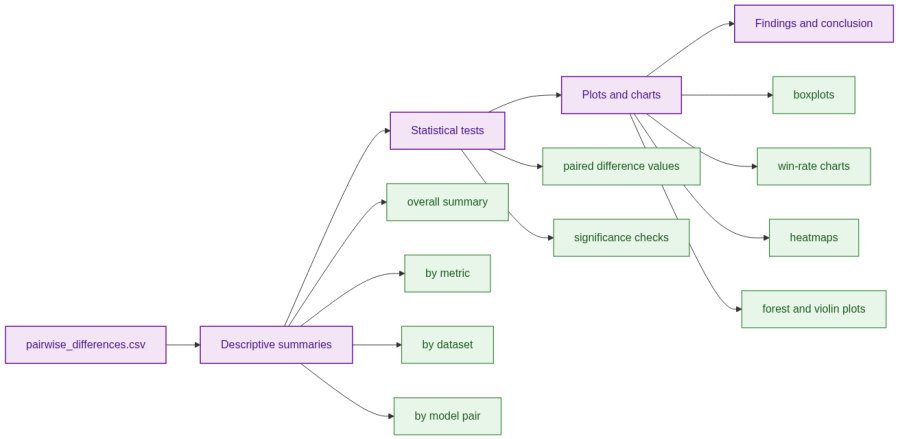
- Experiment organized by task type, dataset, model pair, and split

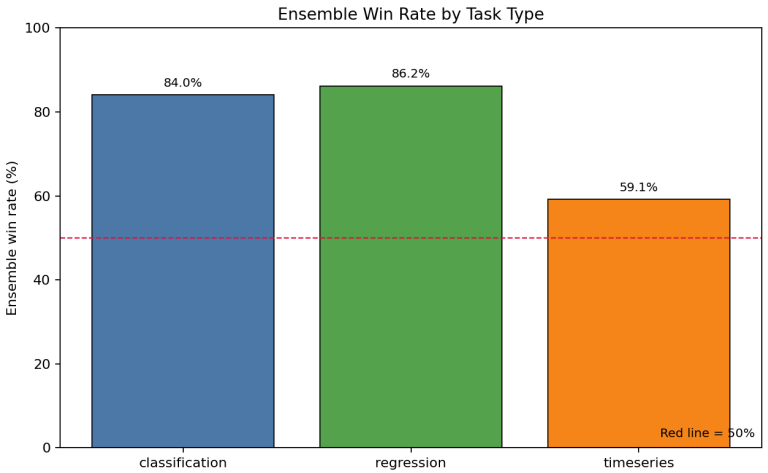
Data Preprocessing



- Same dataset + same split -> matched model comparison row
- Output becomes analysis-ready paired differences

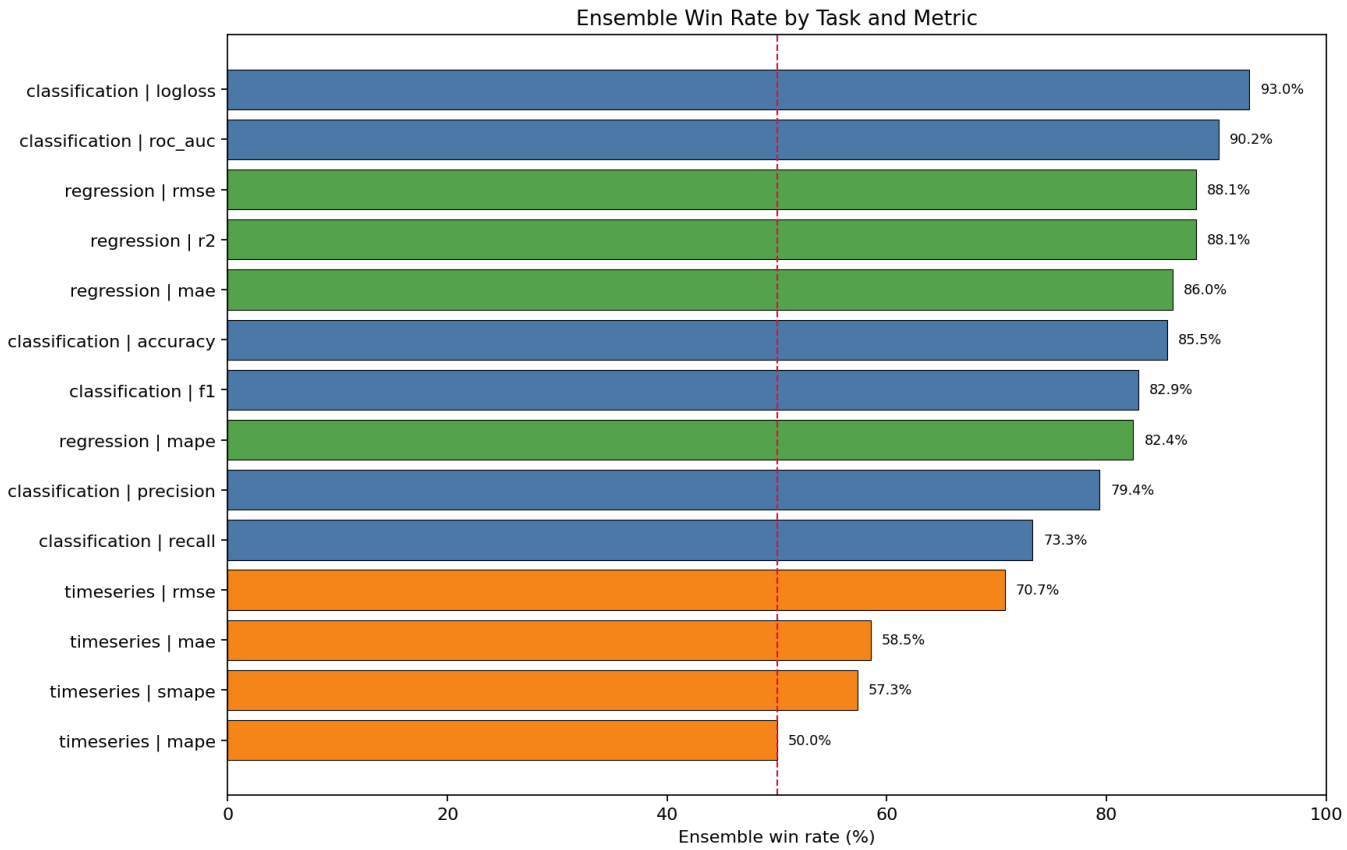
Descriptive Analysis





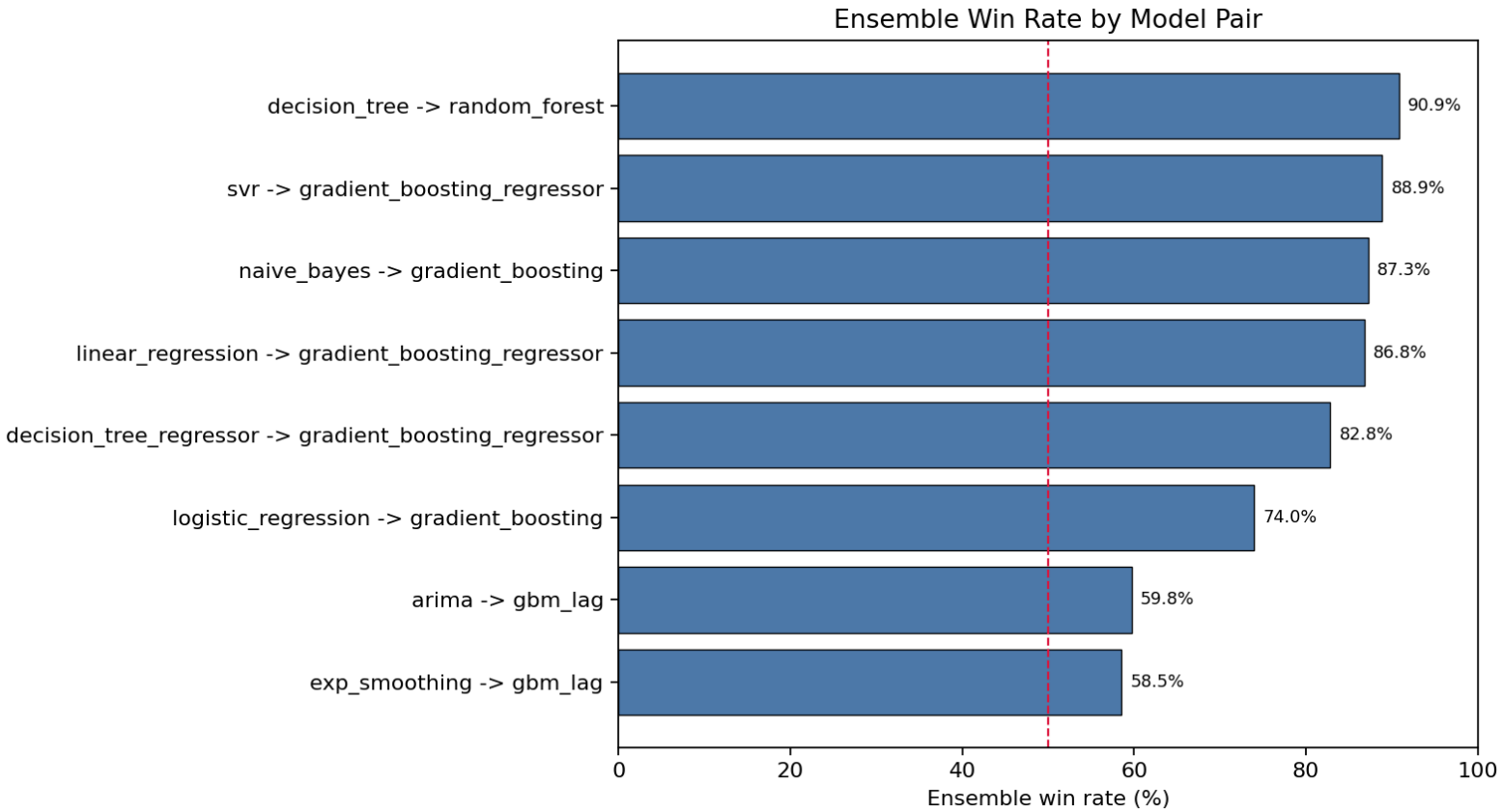
- First ask: where do ensembles win most often?

Inferential Analysis



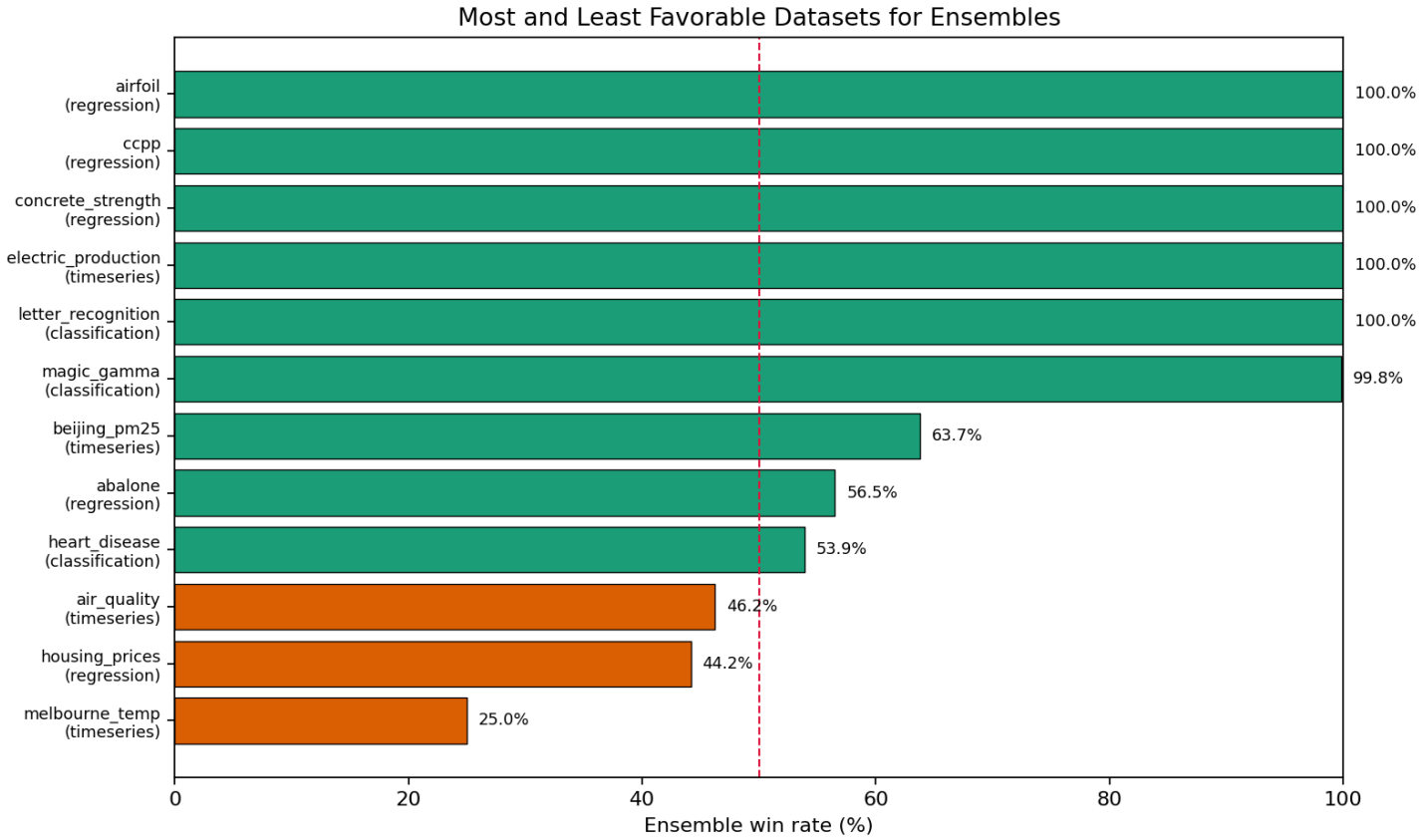
- Paired split structure supports formal significance testing
- Metric view shows whether gains stay stable across measures

Predictive Model



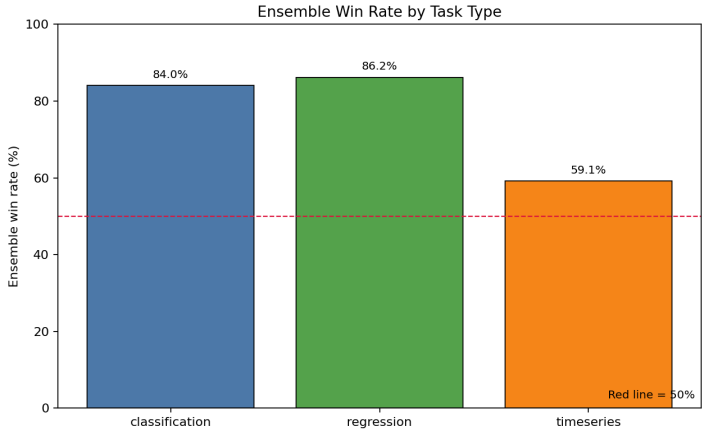
- Compare concrete single -> ensemble upgrades
- Not every ensemble upgrade gives same value

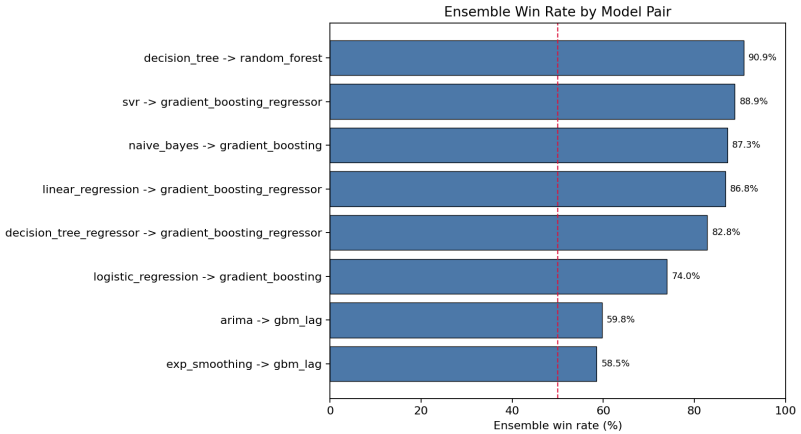
Results



- Best and worst datasets show effect is not universal

Final Decision





- Ensembles usually help in classification and regression
- Final claim must stay conditional on dataset and metric

Conclusion

- Fair pipeline produced both raw runs and paired differences
- Ensembles show strong overall advantage in classification and regression
- Strength of that advantage still depends on dataset, metric, and model pair